Ningchen (Marissa) Ma

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SUMMARY

Software engineer with 3+ years at Amazon building and owning production backend systems, APIs, and GenAI tools at enterprise scale. Yale M.S. Computer Science student with experience in full-stack development and compiler research, focused on scalable systems and AI-powered products that reach real customers.

EDUCATION

Yale University, New Haven, CT — Expected May 2027
M.S., Computer Science
Relevant Coursework: Advanced Deep Learning, FPGA Design, Silicon Compilation, Big Data Systems

McGill University, Montreal, Canada — June 2022
B.S., Computer Science (Honours) | First Class Honours | GPA: 3.87/4.00

SOFTWARE ENGINEERING EXPERIENCE

Software Development Engineer, Amazon, Vancouver, Canada — Oct. 2022 – June 2026
People Experience and Technology (PXT)

- Led backend development for an **enterprise-scale workforce planning system**: APIs, data modeling, testing, deployment, and operations.
- Partnered with **PMs, TPMs, UX designers, and senior leaders**, communicating proactively to keep the product aligned with customer needs.
- Designed an employee-assignment validation algorithm that **prevented inaccurate financial projections**, cutting database reads **~3,000×** across 90,000 employees.
- Designed **40+ GraphQL APIs** for comparing staffing and financial impact across planning scenarios.
- Built a **GenAI analytics assistant and remote MCP server**, cutting data exploration from **minutes to seconds**.
- Developed **React frontend workflows** across multiple internal HR products.

RESEARCH EXPERIENCE

Research Assistant, Shir Group, McGill University, Montreal, Canada — May 2022 – Sept. 2022

- Researched compiler optimization using **equality saturation** and IR selection.
- Built an IR-to-Swift code generator that runs fully connected layers on the **Apple M1 neural accelerator** via MLCompute.

Research Assistant, Cloud/IoT Group, McGill University, Montreal, Canada — Jan. 2022 – Apr. 2022

- Optimized **MQTT communication** on ESP32 devices running RIOT OS for edge computing.
- Deployed RIOT OS to ESP32 and wrote SOPs that other researchers used to extend the protocol.

PROJECT EXPERIENCE

Torchbearer, Hack Nation (3rd Place), Frontend UI & Workflows | GitHub — Oct. 2026

- Built the frontend UI and UX workflows for a voice-assisted tool that turns expert reasoning into interactive work maps and step-by-step guidance for new hires.
- Built a custom **ElevenLabs** voice agent and deployed all infrastructure on **AWS**.
- Set the team's timeline and milestones, and ran **customer interviews** to shape the product.

LEADERSHIP EXPERIENCE

Team Leader, Venture Capital Scout Program — June 2026 – Oct. 2026

- Led a **six-member scout team** through a full startup evaluation and presented the recommendation to the fund's investment reviewers.
- Met founders to evaluate product, market, funding plan, and IP; gathered input from multiple investors to select the final startup.
- Ran weekly check-ins, assigned section owners, and organized in-person team meetups to brainstorm ideas, keeping every member involved.

SKILLS

Programming: Python, Java, JavaScript, C/C++, Scala, OCaml, MIPS Assembly, Bash, HTML/CSS, PHP
Web / AI / APIs: React, GraphQL, REST APIs, GenAI Agents, Model Context Protocol (MCP), AI-assisted development
Cloud: AWS (Lambda, DynamoDB, S3, CDK, etc.)
Collaboration: Cross-functional partnership (PM, TPM, UX, leadership), stakeholder communication